

Toward Agentic Cytology: Integrating Classification and Vision-Language Reporting for Pulmonary Cytology

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Abstract. Although deep learning has enabled accurate classification and vision-language report generation for pulmonary cytology, practical diagnostic workflows still lack an agent capable of integrating multiple AI tools and collaborating with human experts. In this study, we propose an Agentic Cytology Framework that integrates a malignancy classification tool, a cancer subtype classification tool, and a vision-language reporting tool through a large language model (LLM)-based AI agent. The proposed agent evaluates the consistency among tool outputs, automatically generates an integrated report for consistent cases, and escalates inconsistent cases for expert review within a Human-in-the-Loop workflow. Experiments were conducted on a dataset of 1,059 pulmonary cytology images. The malignancy and subtype classification tools achieved accuracies of 98.9% and 81.1%, respectively, while the reporting tool achieved a BLEU-4 score of 0.828. The proposed AI agent automatically generated integrated reports for 89.8% of test cases and escalated only inconsistent cases for expert review. These results demonstrate the feasibility of LLM-based orchestration of multiple diagnostic support tools for pulmonary cytology reporting.

Keywords: Agentic AI, Cytology, Vision-Language Model, Human-in-the-Loop

1 Introduction

Recent advances in deep learning have enabled accurate image classification and vision-language report generation for pulmonary cytology. However, these diagnostic support tools have been developed independently, without verifying the consistency between classification results and generated findings. Consequently, practical diagnostic workflows still require manual validation by experts. Recent advances in Agentic AI have enabled large language models to orchestrate multiple AI tools and support Human-in-the-Loop diagnostic workflows [1]. Therefore, we propose an Agentic Cytology Framework that integrates multiple diagnostic support tools through a large language model (LLM)-based AI agent within a Human-in-the-Loop workflow.

2 Methods

A pulmonary cytology dataset consisting of 1,059 image patches from 279 patients was constructed. Three diagnostic support tools were developed: a malignancy classification tool, a cancer subtype classification tool, and a vision-language reporting tool. The classification models were built using Vision Transformers, while the reporting tool combined an OpenCLIP vision encoder with the GPT-oss large language model. Their outputs were integrated by an LLM-based AI agent, which assessed consistency across the tools, generated an integrated report for consistent cases, and escalated inconsistent cases to expert review within a Human-in-the-Loop workflow.

3 Results

The malignancy and cancer subtype classification tools achieved accuracies of 98.9% and 81.1%, respectively, and the reporting tool achieved a BLEU-4 score of 0.828. Based on these outputs, the proposed AI agent automatically generated integrated reports for 89.8% of the test cases while escalating inconsistent cases for expert review. Representative examples are presented in Fig. 1.

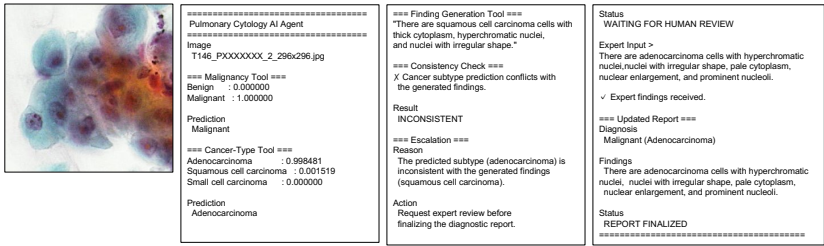


Fig. 1. Representative output of the proposed Agentic Cytology Framework.

4 Conclusion

We proposed an Agentic Cytology Framework that orchestrates multiple diagnostic support tools for pulmonary cytology. By evaluating the consistency among tool outputs, the AI agent automatically generated integrated reports for consistent cases and escalated inconsistent cases for expert review within a Human-in-the-Loop workflow. The proposed framework demonstrates the potential of Agentic AI for reliable clinical diagnostic reporting.

References

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